

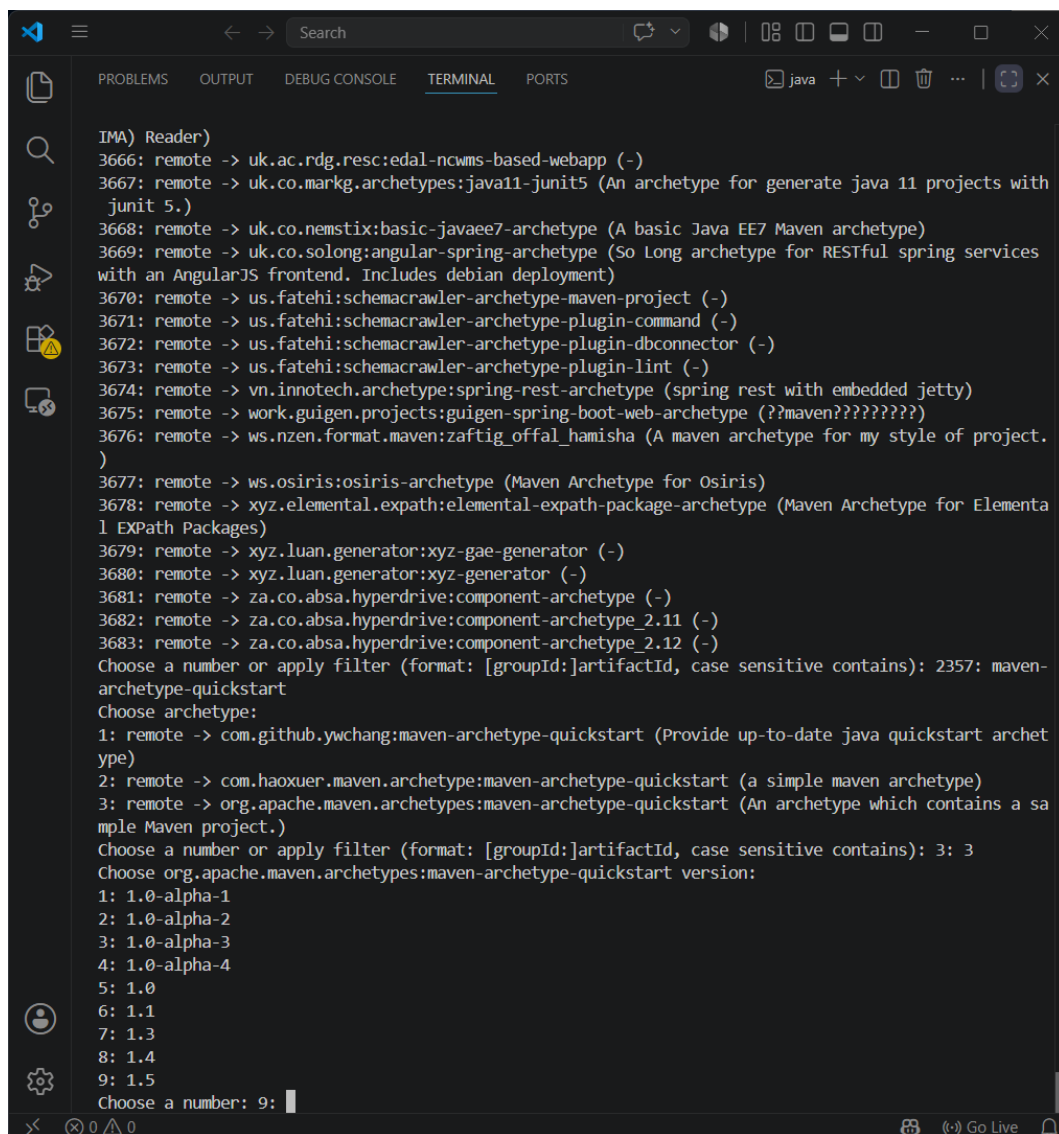
Mockito Exercises

Report Cognizant DN 5.0 | Sanskar Kumawat

Mockito Exercise 1 Report

1. Create a Maven Project

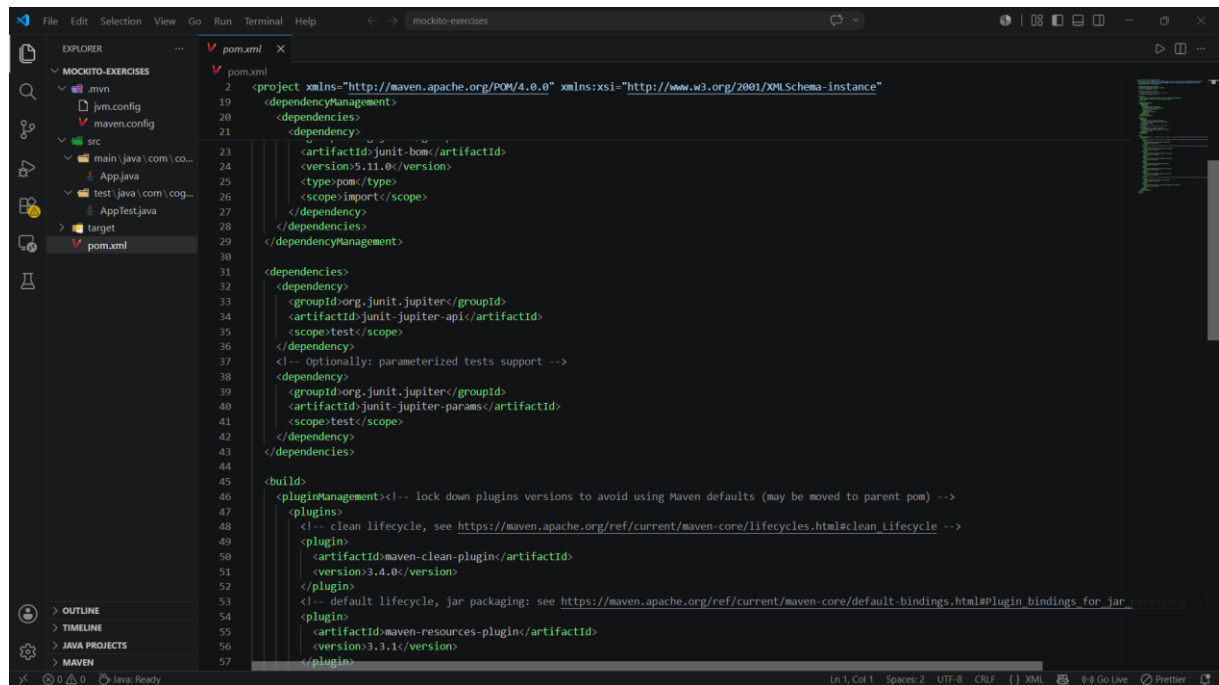
Create a Maven quickstart project (artifact: mockito-exercises).



```
IMA) Reader)
3666: remote -> uk.ac.rdg.resc:edal-ncwms-based-webapp (-)
3667: remote -> uk.co.markg.archetypes:java11-junit5 (An archetype for generate java 11 projects with
junit 5.)
3668: remote -> uk.co.nemstix:basic-javaee7-archetype (A basic Java EE7 Maven archetype)
3669: remote -> uk.co.solong:angular-spring-archetype (So Long archetype for RESTful spring services
with an AngularJS frontend. Includes debian deployment)
3670: remote -> us.fatehi:schemacrawler-archetype-maven-project (-)
3671: remote -> us.fatehi:schemacrawler-archetype-plugin-command (-)
3672: remote -> us.fatehi:schemacrawler-archetype-plugin-dbconnector (-)
3673: remote -> us.fatehi:schemacrawler-archetype-plugin-lint (-)
3674: remote -> vn.innotech.archetype:spring-rest-archetype (spring rest with embedded jetty)
3675: remote -> work.guigen.projects:guigen-spring-boot-web-archetype (??maven????????)
3676: remote -> ws.nzen.format.maven:zaftig_offal_hamisha (A maven archetype for my style of project.
)
3677: remote -> ws.osiris:osiris-archetype (Maven Archetype for Osiris)
3678: remote -> xyz.elemental.expath:elemental-expath-package-archetype (Maven Archetype for Elementa
l EXPath Packages)
3679: remote -> xyz.luan.generator:xyz-gae-generator (-)
3680: remote -> xyz.luan.generator:xyz-generator (-)
3681: remote -> za.co.absa.hyperdrive:component-archetype (-)
3682: remote -> za.co.absa.hyperdrive:component-archetype_2.11 (-)
3683: remote -> za.co.absa.hyperdrive:component-archetype_2.12 (-)
Choose a number or apply filter (format: [groupId:]artifactId, case sensitive contains): 2357: maven-
archetype-quickstart
Choose archetype:
1: remote -> com.github.ywchang:maven-archetype-quickstart (Provide up-to-date java quickstart archet
ype)
2: remote -> com.haoxuer.maven.archetype:maven-archetype-quickstart (a simple maven archetype)
3: remote -> org.apache.maven.archetypes:maven-archetype-quickstart (An archetype which contains a sa
mple Maven project.)
Choose a number or apply filter (format: [groupId:]artifactId, case sensitive contains): 3: 3
Choose org.apache.maven.archetypes:maven-archetype-quickstart version:
1: 1.0-alpha-1
2: 1.0-alpha-2
3: 1.0-alpha-3
4: 1.0-alpha-4
5: 1.0
6: 1.1
7: 1.3
8: 1.4
9: 1.5
Choose a number: 9: 
```

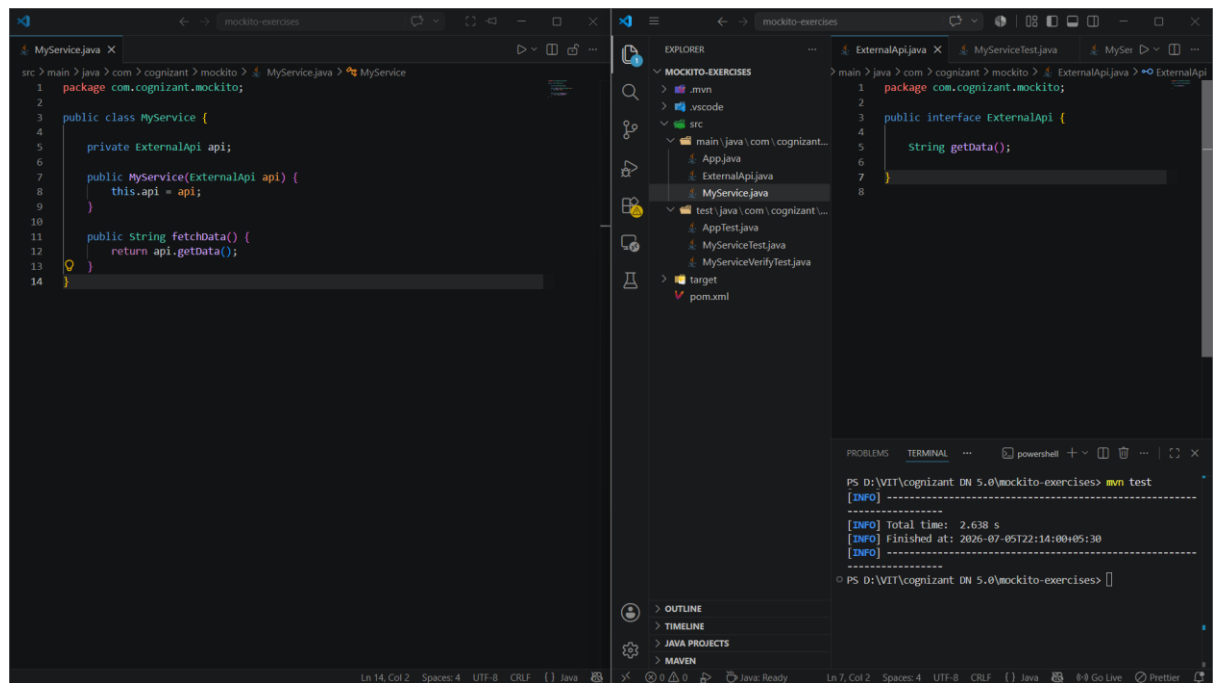
2. Project Structure

Verify the generated Maven project structure and pom.xml.



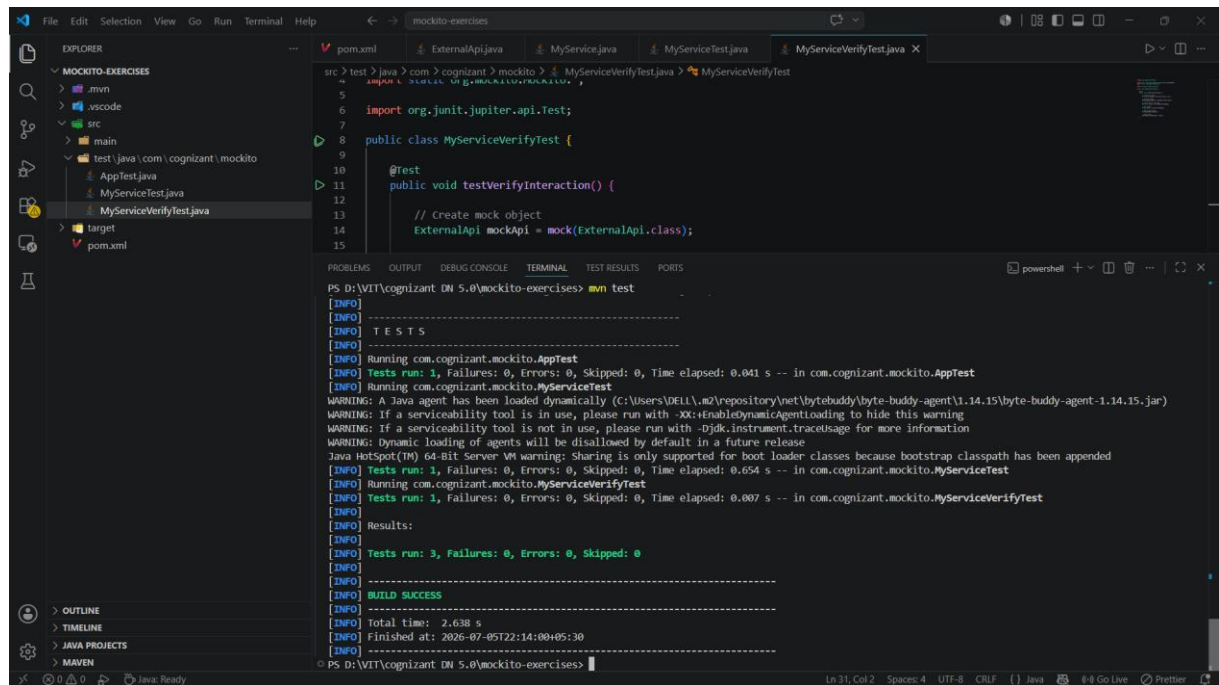
3. Create Java Classes

Create ExternalApi.java, MyService.java and MyServiceTest.java. Add Mockito dependency in pom.xml and write the mock test.



4. Execute the Test

Run 'mvn test'. BUILD SUCCESS confirms the test passed.



The screenshot shows an IDE with a project named 'mockito-exercises'. The file explorer on the left shows the project structure, including 'src/main/java/com/cognizant/mockito' and 'src/test/java/com/cognizant/mockito'. The main editor displays the code for 'MyServiceVerifyTest.java'. The code defines a mock object 'ExternalApi' and a test method 'testVerifyInteraction()' that uses Mockito's 'when' and 'thenReturn' to stub the 'getExternalApi()' method to return the mock object. The terminal at the bottom shows the output of 'mvn test', which includes the Maven version, the test results (3 tests passed, 0 failures, 0 errors, 0 skipped), and the final 'BUILD SUCCESS' message.

```
src > test > java > com > cognizant > mockito > MyServiceVerifyTest.java > MyServiceVerifyTest
4 import static org.mockito.Mockito.*;
5
6 import org.junit.jupiter.api.Test;
7
8 public class MyServiceVerifyTest {
9
10     @Test
11     public void testVerifyInteraction() {
12
13         // Create mock object
14         ExternalApi mockApi = mock(ExternalApi.class);
15
16     }
```

```
PS D:\VIT\cognizant DN 5.0\mockito-exercises> mvn test
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.cognizant.mockito.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.041 s -- in com.cognizant.mockito.AppTest
[INFO] Running com.cognizant.mockito.MyServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.054 s -- in com.cognizant.mockito.MyServiceTest
[INFO] Running com.cognizant.mockito.MyServiceVerifyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.007 s -- in com.cognizant.mockito.MyServiceVerifyTest
[INFO] Results:
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.638 s
[INFO] Finished at: 2026-07-05T22:14:00+05:30
[INFO]
[INFO] -----
```

Key Concepts

- Create a mock using Mockito.
- Stub method behavior with when(...).thenReturn(...).
- Inject the mock into the service.
- Validate the result using assertEquals().
- Execute tests with mvn test.—

EXERCISE 2: Verifying Interactions

In Exercise 2, we check **whether a method was actually called**.

Create a new file: MyServiceVerifyTest.java

```
package com.cognizant.mockito;

import static org.junit.jupiter.api.Assertions.assertEquals;

import static org.mockito.Mockito.*;
```

```

import org.junit.jupiter.api.Test;

public class MyServiceVerifyTest {

    @Test

    public void testVerifyInteraction() {

        ExternalApi mockApi = mock(ExternalApi.class);

        when(mockApi.getData()).thenReturn("Mock Data");

        MyService service = new MyService(mockApi);

        String result = service.fetchData();

        verify(mockApi).getData();

        assertEquals("Mock Data", result);

    }

}

```

```

PS D:\VIT\cognizant DN 5.0\mockito-exercises> mvn test
[INFO] --- compiler:3.13.0:compile (default-compile) @ mockito-exercises ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.3.1:testResources (default-testResources) @ mockito-exercises ---
[INFO] skip non existing resourceDirectory D:\VIT\cognizant DN 5.0\mockito-exercises\src\test\resources
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ mockito-exercises ---
[INFO] Recompiling the module because of added or removed source files.
[INFO] Compiling 3 source files with javac [debug release 17] to target\test-classes
[INFO] --- surefire:3.3.0:test (default-test) @ mockito-exercises ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.cognizant.mockito.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.041 s -- in com.cognizant.mockito.AppTest
[INFO] Running com.cognizant.mockito.MyServiceTest
[INFO] WARNING: A Java agent has been loaded dynamically (C:\Users\DELL\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.15\byte-buddy-agent-1.14.15.jar)
[INFO] WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
[INFO] WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceSage for more information
[INFO] WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.654 s -- in com.cognizant.mockito.MyServiceTest
[INFO] Running com.cognizant.mockito.MyServiceVerifyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.007 s -- in com.cognizant.mockito.MyServiceVerifyTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.638 s
[INFO] Finished at: 2026-07-05T22:14:00+05:30
[INFO]
[INFO]
PS D:\VIT\cognizant DN 5.0\mockito-exercises>

```